EUREKA

STUDY GUIDE

INGENIUM

ROUND 1- RAPID RESEARCH

ROUND OVERVIEW

Delegates take part in a fast-paced, interactive research and analytical assessment delivered on a digital testing platform. The round presents teams with unfamiliar scientific data, claims, and experimental models, requiring rapid evaluation of evidence, identification of methodological flaws, and deduction of valid conclusions under strict time constraints.

The assessment consists of 30 questions distributed across three difficulty tiers: 10 Easy, 10 Medium, and 10 Hard. Success requires balancing analytical rigor with rapid response times across data interpretation, experimental design, and source evaluation.

This is an elimination round. The top-performing teams advance to Round 2.

STRUCTURE OF THE ROUND

Assessment Format: Teams answer 30 timed questions – including multiple-choice, ordering, and data-entry formats – split evenly across three difficulty tiers (10 Easy, 10 Medium, 10 Hard).

Data & Visual Reasoning: Questions assess the ability to process and evaluate visual and tabular data, through data interpretation, graph analysis, graph selection, and experimental anomaly identification.

Critical Evaluation: Questions test the identification of flaws, confounding elements, and source validity, through error detection, source credibility evaluation, correlation vs. causation, and confounding variable analysis.

Scientific Investigation: Questions evaluate the core principles of structured scientific inquiry, through experimental design, research methodology selection, and hypothesis testing.

Evidence & Reasoning: Questions require synthesizing information to validate claims, through evidence matching, inference deduction, conclusion validity, and statistical reasoning.

EVALUATION CRITERIA

Component	Basis
Marking Scheme	Points awarded per correct answer against a predefined answer key; questions weighted by difficulty (Easy / Medium / Hard).
Speed / Tie-Breakers	Response speed on the digital platform automatically calculates score bonuses. If written backup scoring is used, cumulative accuracy plus designated tie-breaker questions determine rank.
Core Competencies	Analytical precision in data analysis, scientific rigor in experimental critique, and deductive logic under time pressure.

DELEGATE CAP

3 delegates per team.

ROUND 2 - SCIENTIFIC CRISIS

ROUND OVERVIEW

Delegates are given a controversial scientific theory or topic at the start of the round. The topic is not revealed beforehand, requiring delegates to quickly develop a scientifically backed and logically sound argument for why the theory is true or false.

Throughout the round, crisis updates are introduced that may either support or contradict the delegate's current position. These updates are treated as 100% factual and accurate. Delegates must critically assess the new information and, where necessary, revise their arguments within the given time limit.

This is an elimination round. The top-performing delegates advance to Round 3.

STRUCTURE OF THE ROUND

Topic Reveal: A controversial scientific theory or topic is revealed at the start of the round.

Initial Research: Delegates research the topic using credible sources and formulate an argument for why the theory is true or false.

Crisis Updates: New pieces of information are introduced throughout the round. Delegates must incorporate these updates into their research and reassess their position where necessary.

Final Submission: Delegates submit their revised research and final argument within the allocated time.

EVALUATION CRITERIA

Criterion	Description
Adaptability	Ability to respond effectively to new information and revise their position.
Scientific Backing	Quality, credibility, and relevance of the evidence used.
Logical Reasoning	Strength, consistency, and validity of the arguments presented.

DELEGATE CAP

3 to 4 delegates per team.

ROUND 3 - EXPERIMENTAL TIER CHALLENGE

ROUND OVERVIEW

The final elimination stage of the Research category. Teams select a difficulty tier and are scored on how they approach and solve an experimental physics problem – not on how they organize themselves internally as a team.

STRUCTURE OF THE ROUND

Stage 1 – Tier Commitment: Teams pre-commit to a tier (Easy / Medium / Hard) before the round starts, not after seeing the problem. If a tier fills up, priority goes to teams with the stronger combined standing from Rounds 1–2. Once committed, there is no switching.

Stage 2 – Problem Reveal: Sealed problem cards are opened simultaneously across all stations, all tiers together, on the same start signal – regardless of tier. This keeps the round a genuine gamble rather than an easier ride on the same clock.

Stage 3 – Design & Execution Window: Every team, in every tier, works within the identical time window to design, run, and prepare their answer. Higher tiers pack more conceptual complexity and more open-ended design decisions into that same time window.

Stage 4 – Submission: Once the window closes, teams finalize a standardized write-up (setup/approach, data, calculation, final result, conclusion), keeping judging consistent no matter which tier a team ran.

Stage 5 – Judging: Judging takes place after the round has finished, not during it. Once every team has completed their experiment and prepared their answer, judges go team to team to evaluate approach, methodology, and result.

EVALUATION CRITERIA

Scoring weighs reasoning and method heavily, not just whether the final result lands close to expected – this matters more at higher tiers, where the “right” approach is not obvious. Judging is scoped entirely to the team's problem-solving approach and execution, not to how tasks were divided internally.

DELEGATE CAP

3 delegates per team.